

Round and square FRP cooling towers

WCT RL / WCT SL series

Compare round and square evaporative towers, with separate model tables for each shape.



Capacity	Outlet water	Inlet water temperature	Inlet water pressure
10 to 300 TR	Wet bulb + 4 °C	40 to 100 °C	2 to 3 bar g

Inside this catalogue

Round FRP Cooling Towers models	2	Square FRP Cooling Towers models	3
How it works & construction	4	Selection factors	5
Applications & selection questions	6		



Specifications online

Layout edition 2026-10 · Product-range information. Confirm the selected model and duty with Win Equipments before ordering.

Round FRP Cooling Towers models

Rating conditions

Fills	L: up to 55 °C inlet · H: 55–85 °C inlet · P: polypropylene rings	Motor	TEFC IP55, 415 V / 3 Ph / 50 Hz
Working pressure	2 to 3 bar g	Evaporation loss	$0.00085 \times 1.8 \times \text{flow} \times \Delta T$

Capacity & operating data

Model	Capacity TR	Water flow m ³ /hr	Motor HP	Motor speed rpm	Fan diameter mm
WCT 010 RL	10	6	0.5	1440	450
WCT 020 RL	20	12	1	960	750
WCT 030 RL	30	18	1	960	750
WCT 040 RL	40	24	1	960	850
WCT 050 RL	50	30	2	960	900
WCT 060 RL	60	36	3	960	1160
WCT 080 RL	80	48	3	960	1160
WCT 100 RL	100	60	5	960	1400
WCT 125 RL	125	75	5	960	1400
WCT 150 RL	150	90	5	960	1600
WCT 200 RL	200	120	7.5	710	1800
WCT 300 RL	300	180	12.5	600	2400

Construction, dimensions & connections

Model	In / out NB	Dia × H mm	Shipping weight kg	Operating weight kg
WCT 010 RL	1½" / 1½"	930 × 1690	80	280
WCT 020 RL	2" / 2"	1380 × 1690	100	450
WCT 030 RL	3" / 3"	1660 × 1860	175	540
WCT 040 RL	3" / 3"	1780 × 1950	205	700
WCT 050 RL	3" / 3"	1990 × 2120	260	760
WCT 060 RL	3" / 3"	1920 × 2720	330	800
WCT 080 RL	4" / 4"	2150 × 2670	375	890
WCT 100 RL	5" / 5"	2895 × 3050	615	1850
WCT 125 RL	5" / 5"	2950 × 3250	805	2300
WCT 150 RL	5" / 5"	3000 × 3950	1005	2650
WCT 200 RL	6" / 6"	3300 × 4230	1110	3050
WCT 300 RL	8" / 8"	4450 × 3900	1950	4050

- Model code: WCT + capacity in TR + R (round) + fill type: L low-temperature fills (up to 55 °C inlet), H high-temperature fills (55–85 °C inlet), P polypropylene rings. Working pressure 2–3 bar g max.
- Each model is available in L, H and P variants.
- Rated power is the maximum power consumed at rated conditions.

Square FRP Cooling Towers models

Rating conditions

Fills	L: up to 55 °C inlet · H: 55–85 °C inlet · P: polypropylene rings	Motor	TEFC IP55, 415 V / 3 Ph / 50 Hz
Drift eliminator	Rigid PVC, 600 × 600 × 130 mm	Evaporation loss	$0.00085 \times 1.8 \times \text{flow} \times \Delta T$

Capacity & operating data

Model	Capacity TR	Water flow m³/hr	Motor HP	Motor speed rpm	Fan diameter mm
WCT 010 SL	10	6	0.5	1440	450
WCT 020 SL	20	12	0.5	1440	450
WCT 030 SL	30	18	1	960	750
WCT 040 SL	40	24	1	960	800
WCT 060 SL	60	36	2	960	1160
WCT 100 SL	100	60	5	960	1400
WCT 150 SL	150	90	7.5	960	1400
WCT 200 SL	200	120	7.5	710	1800
WCT 250 SL	250	150	10	710	1800
WCT 300 SL	300	180	12.5	600	2200

Construction, dimensions & connections

Model	In / out NB	L × D × H mm	Shipping weight kg	Operating weight kg
WCT 010 SL	2" / 2"	825 × 825 × 1500	85	210
WCT 020 SL	2" / 2½"	825 × 825 × 1900	90	225
WCT 030 SL	2" / 2½"	1000 × 1000 × 2100	240	750
WCT 040 SL	2½" / 2½"	1170 × 1170 × 2600	320	850
WCT 060 SL	3" / 3"	1400 × 1400 × 2400	350	910
WCT 100 SL	5" / 5"	1750 × 1750 × 2800	525	1550
WCT 150 SL	6" / 6"	2400 × 2400 × 3100	700	2100
WCT 200 SL	8" / 8"	2400 × 2400 × 3300	775	2250
WCT 250 SL	8" / 8"	3000 × 3000 × 3200	1110	3100
WCT 300 SL	8" / 8"	3000 × 3000 × 3200	1250	3300

- Model code: WCT + capacity in TR + S (square) + fill type: L low-temperature fills (up to 55 °C inlet), H high-temperature fills (55–85 °C inlet), P polypropylene rings. Working pressure 2–3 bar g max.
- Each model is available in L, H and P variants.
- Rated power is the maximum power consumed at rated conditions.

How it works & construction

How it works · Round FRP Cooling Towers

A cooling tower is a heat-rejection device that extracts waste heat from the application to the atmosphere using a fan and motor. The heat rejection is evaporative.

1 Distribute

Hot water enters the main header and is distributed through the rotary sprinkler, whose jet recoil spins it for uniform flow.

3 Evaporate

The axial fan draws air up through the fills; a small part of the water evaporates and cools the rest.

2 Film

Water spreads over the honeycomb PVC fills as a film with a large contact surface.

4 Collect

Cooled water falls as droplets into the basin and is pumped back to the equipment.

Round FRP Cooling Towers: construction

FRP casing

- Gel-coated fibre glass reinforced polyester
- Compact and light for easy installation

Rotary sprinkler

- Jet recoil drives rotation for even distribution
- Nylon and aluminium construction
- Minimises drift loss

Fan

- Dynamically balanced axial type
- Aluminium alloy and Nylon 66
- Low noise, low running cost

PVC fills

- Rigid PVC honeycomb, very large contact area
- 600 × 300 × 150 mm blocks
- Flute height 12–19 mm, sheet 0.2–0.25 mm

Motor

- Squirrel cage induction, TEFC, IP55
- Flanged with extended shaft
- Hindustan / Stark make

PP rings option

- Easy to clean
- Withstands more than 90 °C
- Rigid, individual, not bonded

Square FRP Cooling Towers: construction

FRP casing

- Gel-coated fibre glass reinforced polyester
- Compact and light for easy installation

Drift eliminators

- Rigid PVC, 600 × 600 × 130 mm
- Reduce carry-over loss of water

Motor & fan

- TEFC IP55 squirrel cage motor, extended shaft
- Dynamically balanced axial fan in aluminium alloy and Nylon 66

Spray nozzles

- Even distribution through a wide spray angle
- Used on square towers

PVC fills

- Rigid PVC honeycomb, very large contact area
- Flute height 12–19 mm

Selection factors

Selection factors · Round FRP Cooling Towers

Selection factor by cooling range (ΔT) and approach to wet bulb, round towers.

Range ΔT (°C) / approach (°C)	2	3	4	5	6	7	8
4	10	11	12	13	14	15	16
6	9	10	11	12	13	14	15
8	8	9	10	11	12	13	14
10	7	8	9	10	11	12	13
12	6	7	8	9	10	11	12

$$\text{Evaporation loss} = 0.00085 \times 1.8 \times \text{flow rate (m}^3/\text{hr)} \times \Delta T$$

Selection factors · Square FRP Cooling Towers

Selection factor by cooling range (ΔT) and approach to wet bulb, square towers.

Range ΔT (°C) / approach (°C)	4	5	6	7	8
4	13	14	15	16	17
6	12	13	14	15	16
8	11	12	13	14	15
10	10	11	12	13	14
12	9	10	11	12	13
14	8	9	10	11	12
16	7	8	9	10	11

$$\text{Evaporation loss} = 0.00085 \times 1.8 \times \text{flow rate (m}^3/\text{hr)} \times \Delta T$$

Applications & selection questions

Round FRP Cooling Towers

Applications: Automobile · Cement · Chemical processing · Diesel generators · Distilleries & breweries · Soft drinks · Electronics · Food & beverage · Glass · Health care & hospitals · Induction furnaces · Paper mills · Plastic machinery · PET bottle machines · Spinning, knitting & hosiery · Sugar

What outlet water temperature can a WCT tower achieve?

Outlet water is rated at wet bulb temperature + 4 °C.

Which fill should I choose?

L (low-temperature) fills for inlet water up to 55 °C, H (high-temperature) fills for 55–85 °C, and polypropylene rings (P) for hot or dirty water up to above 90 °C.

How much water is lost to evaporation?

Evaporation loss $\approx 0.00085 \times 1.8 \times \text{flow rate (m}^3/\text{hr)} \times \Delta T$. For example, 30 m³/hr with a 5 °C range loses about 0.23 m³/hr.

Where should the tower be installed?

Install the tower so the pump sits below the sump outlet level; never place the pump above the sump outlet.

Square FRP Cooling Towers

Applications: Automobile · Cement · Chemical processing · Diesel generators · Food & beverage · Glass · Hospitals · Induction furnaces · Paper mills · Plastic machinery · PET bottle machines · Textile mills · Sugar

Round or square tower: which should I choose?

Both use the same evaporative principle and capacity range. Square towers suit tight plots, rooftops and installations against walls; round towers use a rotary sprinkler instead of spray nozzles.

What outlet water temperature can I expect?

Outlet water is rated at wet bulb temperature + 4 °C.

Which fill should I choose?

L fills for inlet water up to 55 °C, H fills for 55–85 °C, and polypropylene rings (P) for hot or dirty water.

Confirm your selection

Send the model, required capacity, operating temperatures, pressure and site conditions. For a model-specific datasheet or GA drawing, include your installation requirement.

Win Equipments

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Catalogue values are for product selection. Confirm duty, materials, connections and final dimensions with Win Equipments before ordering. Layout edition 2026-10.